Time Series Cheat Sheet

By Marcelo Moreno Porras - Universidad Rey Juan Carlos
The Econometrics Cheat Sheet Project

Basic concepts

Definitions

Time series: Succession of observations ordered in time with a fixed frequency.

Given the format of a time series:

- **Point-in-time (stock)**: A single value is recorded for each period.
- **Aggregated (flow)**: Values represent totals or averages over the period.
- **Range/interval (OHLC)**: Each period records multiple summary statistics, such as min, max, open, close.

Stochastic process: A sequence of random variables that are indexed in time.

Components of a time series

- **Trend**: The long-term general movement of a series.
- **Seasonal variations**: Periodic oscillations that are produced in a period less than or equal to a year, and can be easily identified across different years (usually the result of climatology).
- **Cyclical variations**: Periodic oscillations that are produced in a period greater than a year (are the result of the economic cycle).
- **Residual variations**: Movements that do not follow a recognizable periodic oscillation (irregular events).

Type of time series models

- **Static models**: The relation between y and x is contemporaneous. Conceptually:
$$y_t = \beta_0 + \beta_1 x_t + u_t$$
- **Distributed-lag models**: The relation between y and x is not contemporaneous. Conceptually:
$$y_t = \beta_0 + \beta_1 x_t + \beta_2 x_{t-1} + \cdots + \beta_s x_{t-s} + u_t$$
The long-term cumulative effect in y when $\Delta x = 1$ is:
$$\beta_1 + \beta_2 + \cdots + \beta_s$$
- **Dynamic models**: Introduces lags of the dependent variable. Conceptually:
$$y_t = \beta_0 + \beta_1 y_{t-1} + \cdots + \beta_s y_{t-s} + u_t$$
- Combinations of the above, like the rational distributed-lag models (distributed-lag + dynamic).

Assumptions and properties

OLS assumptions in time series

Gauss-Markov assumptions extended for time series:

t1. **Linearity in parameters and weak dependence**:
   a. y_t must be a linear function of the β's.
   b. The stochastic $\{(x_t, y_t) : t = 1, 2, \ldots, T\}$ is stationary and weakly dependent.

t2. **No perfect collinearity**:
   - There are no independent variables that are constant: $\text{Var}(x_{jt}) \neq 0 \quad \forall j = 1, \ldots, k$
   - There is no exact linear relation between independent variables.

t3. **Conditional mean zero and correlation zero**:
   a. There are no relevant variables left out of the model: $\text{Cov}(x_{jt}, u_t) = 0 \quad \forall j = 1, \ldots, k \rightarrow$ **exogeneity**
   b. There are no systematic errors: $\text{E}(u_t \mid \text{X}) = \text{E}(u_t \mid \ldots, \text{x}_{t+1}, \text{x}_t, \text{x}_{t-1}, \ldots) = 0 \quad \forall t \rightarrow$ **strict exogeneity** (3a holds)

t4. **Homoscedasticity**: The variability of the error term is the same for all levels of x: $\text{Var}(u_t \mid x_{1t}, \ldots, x_{kt}) = \sigma^2 \quad \forall t = 1, \ldots, T$

t5. **No autocorrelation**: Errors do not contain information about any other errors: $\text{Corr}(u_t, u_s \mid x_1, \ldots, x_k) = 0 \quad \forall t \neq s$

t6. **Normality**: Errors are independent and identically distributed (**i.i.d.**): $u_t \sim \mathcal{N}(0, \sigma^2)$

t7. **Data size**: The number of observations available must be greater than $(k + 1)$ parameters to estimate. (It is already satisfied under asymptotic situations)

Asymptotic properties of OLS

Under the Gauss-Markov assumptions and the Central Limit Theorem (CLT):

- Hold t1 to t3a: OLS is **consistent**. $\text{plim}(\hat{\beta}_j) = \beta_j$
- Hold t1 to t3b: OLS is **unbiased**. $\text{E}(\hat{\beta}_j) = \beta_j$
- Hold t1 to t5: **Asymptotic normality** of estimators: $\hat{\beta}_j \underset{a}{\sim} \mathcal{N}(\beta_j, \text{Var}(\hat{\beta}_j))$
- Hold t1 to t5: **Unbiased estimate** of σ^2. $\text{E}(\hat{\sigma}^2) = \sigma^2$
- Hold t1 to t5: OLS is BLUE (Best Linear Unbiased Estimator) or **efficient**.
- Hold t1 to t6: Hypothesis testing and confidence intervals can be done reliably.

Spurious regression: Is when a false statistically significant relationship is found between independent non-stationary time series (e.g., due to shared trends), despite having no true causal link. This **breaks the first (t1b)** OLS **assumption**.

Trends and seasonality

Trends

Two time series can have the same (or contrary) trend, which should lead to a high level of correlation. This can provoke a false appearance of causality; the problem is **spurious correlation**. Given the model:

$$y_t = \beta_0 + \beta_1 x_t + u_t$$

where:

$$y_t = \alpha_0 + \alpha_1 \text{Trend} + v_t$$
$$x_t = \gamma_0 + \gamma_1 \text{Trend} + w_t$$

Adding a trend to the model can solve the problem:

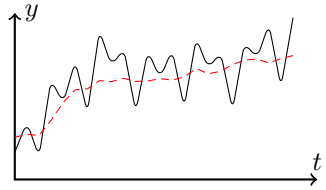
$$y_t = \beta_0 + \beta_1 x_t + \beta_2 \text{Trend} + u_t$$

The trend can be linear or non-linear (quadratic, cubic, exponential, etc.)

Another way is to make use of the **Hodrick-Prescott filter** to extract the trend and the cyclical component.

Seasonality

A time series can manifest seasonality. That is, the series is subject to seasonal variations or patterns, usually related to the climate.



For example, GDP is usually higher in summer and lower in winter (seasonally adjusted series in dashed red).

- Regressing time series that present seasonality can lead to **spurious results**.

There are different **seasonal adjustment** methods:

a. Include seasonal binary variables in the model. For example, for quarterly series (Sq_t are binary variables):
$$y_t = \beta_0 + \beta_1 S2_t + \beta_2 S3_t + \beta_3 S4_t + \beta_4 x_{1t} + \cdots + \beta_k x_{kt} + u_t$$
b. Seasonally adjust the variables and then perform the regression with the adjusted variables.
c. Take seasonal differences (s is the seasonal period):
$$\Delta_s y_t = y_t - y_{t-s}$$
d. Apply **X-13ARIMA-SEATS** adjustment.

Autocorrelation

The error terms u_t and u_s are correlated across different time periods conditional on the explanatory variables. This **breaks the fifth (t5) OLS assumption**.

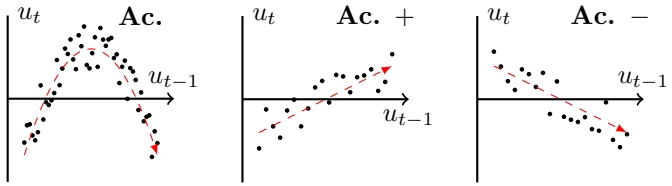
$$\text{Corr}(u_t, u_s \mid x_1, \ldots, x_k) \neq 0 \quad \exists t \neq s$$

Consequences

- OLS estimators are still consistent.
- OLS estimators are still unbiased.
- OLS is **no longer efficient**, but still a LUE (Linear Unbiased Estimator).
- **Variance estimations** of the estimators are **biased**: the construction of confidence intervals and the hypothesis testing is not reliable.

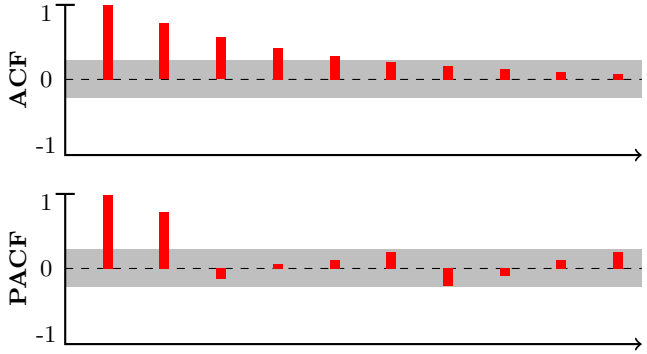
Detection

Scatter plots: Identify scatter patterns on u_{t-1} vs. u_t.



Correlogram: Autocorrelation function (ACF) and partial ACF (PACF).

- Y axis: correlation.
- X axis: lag number.
- Grey area: $\pm 1.96/T^{0.5}$



- **MA**(q) **process**. ACF: only the first q coefficients are significant, the remaining are abruptly cancelled. PACF: attenuated exponential fast decay or sine waves.
- **AR**(p) **process**. ACF: attenuated exponential fast decay or sine waves. PACF: only the first p coefficients are significant, the remaining are abruptly cancelled.
- **ARMA**(p, q) **process**. ACF and PACF: the coefficients are not abruptly cancelled and present a fast decay.

If the ACF coefficients do not decay rapidly, there is a clear indicator of a lack of stationarity in mean.

Formal tests:

Supposing that u_t follows an AR(1) process:

$$u_t = \rho_1 u_{t-1} + \varepsilon_t$$

where ε_t is white noise.

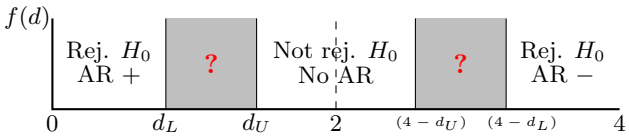
- **AR(1) t test**:
  - Exogenous regressors.
  - H_0: No autocorrelation.
  - H_1: Autocorrelation of order one, AR(1).

$$\text{Under } H_0: \quad t = \frac{\hat{\rho}_1}{\text{se}(\hat{\rho}_1)} \sim t_{T-k-1}$$

- **Durbin-Watson statistic**:
  - Exogenous regressors and residual normality.
  - H_0: No autocorrelation.
  - H_1: Autocorrelation of order one, AR(1).

$$\text{Under } H_0: \quad d = \frac{\sum_{t=2}^{T}(\hat{u}_t - \hat{u}_{t-1})^2}{\sum_{t=1}^{T}\hat{u}_t^2} \approx 2 \cdot (1 - \hat{\rho}_1)$$

where $0 \leq d \leq 4$.



- **Durbin's h**:
  - Dynamic models.
  - H_0: No autocorrelation.
  - H_1: Autocorrelation of order one, AR(1).

$$\text{Under } H_0: \quad h = \hat{\rho} \cdot \sqrt{\frac{T}{1 - T \cdot \widehat{\text{Var}}(\hat{\beta}_{\text{lag}})}} \underset{a}{\sim} \mathcal{N}(0,1)$$

- **Breusch-Godfrey test**:
  - MA(q): $u_t = \varepsilon_t - m_1\varepsilon_{t-1} - \cdots - m_q\varepsilon_{t-q}$
  - AR(p): $u_t = \rho_1 u_{t-1} + \cdots + \rho_p u_{t-p} + \varepsilon_t$
  - H_0: No autocorrelation.
  - H_1: Autocorrelation of order q or p.

$$\text{Under } H_0: \quad T \cdot R^2_{\hat{u}_t} \underset{a}{\sim} \chi^2_{q \text{ or } p}$$

- **Ljung-Box Q test**:
  - H_0: No autocorrelation up to lag h.
  - H_1: At least one autocorrelation up to lag h.

$$\text{Under } H_0: \quad Q_{LB} = T(T+2)\sum_{k=1}^{h}\frac{\hat{\rho}_k^2}{T-k} \underset{a}{\sim} \chi^2_h$$

Correction

- Use OLS with a variance-covariance matrix estimator that is **robust to heteroscedasticity and autocorrelation** (HAC), for example, the one proposed by **Newey-West**.
- Use **Generalized Least Squares** (GLS). Supposing $y_t = \beta_0 + \beta_1 x_t + u_t$, with $u_t = \rho u_{t-1} + \varepsilon_t$, where $|\rho| < 1$ and ε_t is white noise.
  - If ρ is **known**, use a **quasi-differentiated model**:

$$y_t - \rho y_{t-1} = \beta_0(1-\rho) + \beta_1(x_t - \rho x_{t-1}) + u_t - \rho u_{t-1}$$
$$y_t^* = \beta_0^* + \beta_1' x_t^* + \varepsilon_t$$

    where $\beta_1' = \beta_1$; and estimate it by OLS.
  - If ρ is **not known**, estimate it by—for example—the **Cochrane-Orcutt iterative method**:
    1. Obtain $\hat{u}_t$ from the original model.
    2. Estimate $\hat{u}_t = \rho\hat{u}_{t-1} + \varepsilon_t$ and obtain $\hat{\rho}$.
    3. Create a quasi-differentiated model:

$$y_t - \hat{\rho} y_{t-1} = \beta_0(1-\hat{\rho}) + \beta_1(x_t - \hat{\rho} x_{t-1}) + u_t - \hat{\rho} u_{t-1}$$
$$y_t^* = \beta_0^* + \beta_1' x_t^* + \varepsilon_t$$

    where $\beta_1' = \beta_1$; and estimate it by OLS.
    4. Obtain $\hat{\beta}_0 = \hat{\beta}_0^*/(1-\hat{\rho})$
    5. Compute $\hat{u}_t = y_t - (\hat{\beta}_0 + \hat{\beta}_1' x_t)$.
    6. Repeat from step 2. The algorithm ends when the estimated parameters vary very little between iterations.
- If not solved, look for **high dependence** in the series.

Exponential smoothing

Given $\{y_t\}$, the smoothed series $\{f_t\}$:

$$f_t = \alpha y_t + (1-\alpha) f_{t-1}$$

where $0 < \alpha < 1$ is the smoothing factor and $f_0 = y_0$.

Forecasts

Two types of forecasts:

- Of the mean value of y for a specific value of x.
- Of an individual value of y for a specific value of x.

Theil's U statistic: Compares the forecast results with the ones of forecasting with minimal historical data.

$$U = \sqrt{\frac{\sum_{t=1}^{T-1}\left(\frac{\hat{y}_{t+1} - y_{t+1}}{y_t}\right)^2}{\sum_{t=1}^{T-1}\left(\frac{y_{t+1} - y_t}{y_t}\right)^2}}$$

- < 1: The forecast is better than guessing.
- $= 1$: The forecast is about as good as guessing.
- > 1: The forecast is worse than guessing.

Stationarity

Stationarity allows to correctly identify relations—that stay unchanged with time—between variables.

- **Stationary process** (strict stationarity): The joint probability distribution of the process remains unchanged when shifted h periods.
- **Non-stationary process**: For example, a series with trend, where at least the mean changes with time.
- **Covariance stationary process**: It is a weaker form of stationarity:
  - $\mathrm{E}(x_t)$ is constant.
  - $\mathrm{Var}(x_t)$ is constant.
  - For any $t, h \geq 1$, $\mathrm{Cov}(x_t, x_{t+h})$ depends only on h, not on t.

Weak dependence

Weak dependence replaces the random sampling assumption for time series.

- A stationary process $\{x_t\}$ is **weakly dependent** when x_t and x_{t+h} are almost independent as h increases without a limit.
- A covariance stationary process is **weakly dependent** if the correlation between x_t and x_{t+h} tends to 0 fast enough when $h \to \infty$ (they are not asymptotically correlated).

Most weakly dependent stationary processes are **integrated of order zero**, I(0). Some examples:

- **Moving average**: $\{x_t\}$ is a moving average of order q, MA(q):
  $$x_t = e_t + m_1 e_{t-1} + \cdots + m_q e_{t-q}$$
  where $\{e_t : t = 0, 1, \ldots, T\}$ is an *i.i.d.* sequence with zero mean and σ_e^2 variance.
- **Autoregressive process**: $\{x_t\}$ is an autoregressive process of order p, AR(p):
  $$x_t = \rho_1 x_{t-1} + \cdots + \rho_p x_{t-p} + e_t$$
  where $\{e_t : t = 1, 2, \ldots, T\}$ is an *i.i.d.* sequence with zero mean and σ_e^2 variance.
  Stability condition: If all roots z of the characteristic equation $1 - \rho_1 z - \cdots - \rho_p z^p = 0$ satisfy $|z| > 1$, then $\{x_t\}$ is an AR(p) stable process that is weakly dependent. For AR(1), the condition is: $|\rho_1| < 1$.
- **ARMA process**: Is a combination of the previous; $\{x_t\}$ is an ARMA(p, q):
  $x_t = e_t + m_1 e_{t-1} + \cdots + m_q e_{t-q} + \rho_1 x_{t-1} + \cdots + \rho_p x_{t-p}$

Unit roots

A process is integrated of order d, I(d), if applying differences d times makes the process stationary.
When $d \geq 1$, the process is said to have a **unit root**. A process has a unit root when the stability condition is not met (there are roots on the unit circle).

Strong dependence

Generally, economic series are highly persistent in time. Some examples of **unit root** I(1):

- **Random walk**: An AR(1) process with $\rho_1 = 1$.
  $$y_t = y_{t-1} + e_t$$
  where $\{e_t : t = 1, 2, \ldots, T\}$ is an *i.i.d.* sequence with zero mean and σ_e^2 variance.
- **Random walk with a drift**: An AR(1) process with $\rho_1 = 1$ and a constant.
  $$y_t = \beta_0 + y_{t-1} + e_t$$
  where $\{e_t : t = 1, 2, \ldots, T\}$ is an *i.i.d.* sequence with zero mean and σ_e^2 variance.

Unit root tests

Test	H_0	Reject H_0
ADF	I(1)	$\tau <$ Critical value
KPSS	I(0) level	$\eta_\mu >$ Critical value
KPSS	I(0) trend	$\eta_\tau >$ Critical value
Phillips-Perron	I(1)	$\mathrm{Z}(\tau) <$ Critical value
Zivot-Andrews*	I(1)	$t_{\min} <$ Critical value

* Tests for unit root in the presence of structural breaks.

From unit root to weak dependence

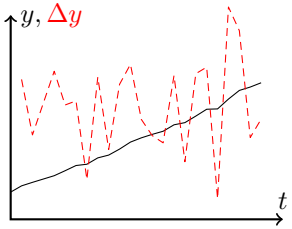
Integrated of **order one**, I(1), means that **the first difference** of the process is **weakly dependent** or I(0) (and usually, stationary). Let $\{y_t\}$ be a random walk:

$$\Delta y_t = y_t - y_{t-1} = e_t$$

where $\{e_t\} = \{\Delta y_t\}$ is *i.i.d.*

Note:

- The first difference often removes trends.
- Logarithms often stabilize the variance.



When an I(1) series is strictly positive, logs are often used before differencing to approximate percentage changes:

$$\Delta \log(y_t) = \log(y_t) - \log(y_{t-1}) \approx \frac{y_t - y_{t-1}}{y_{t-1}}$$

Ergodicity

A strictly stationary process $\{y_t\}$ is **ergodic** if time averages converge to their ensemble averages (expectations). This is often ensured by **strong mixing**, which implies asymptotic independence of distant events.

$$\frac{1}{T}\sum_{t=1}^{T} y_t \underset{a}{\to} \mathrm{E}(y_t)$$

Without it, sample moments may not reflect population moments. Estimators are inconsistent.

Cointegration

Two I(1) series are **cointegrated** if a linear combination is I(0). In that case, a regression between them is not spurious but reflects a valid **long-run** relationship. Cointegrated variables share a common stochastic trend.
For example, $\{x_t\}$ and $\{y_t\}$ are I(1), but $y_t - \beta x_t = u_t$ where $\{u_t\}$ is I(0). (β is the cointegrating parameter).

Cointegration test

Engle-Granger test:

1. Estimate $y_t = \alpha + \beta x_t + u_t$ and obtain $\hat{u}_t$.
2. Perform an ADF test on $\hat{u}_t$ with a modified distribution. The result of this test is equivalent to:
   - H_0: $\hat{u}_t \sim$ I(1) $\to$ No cointegration exists.
   - H_1: $\hat{u}_t \sim$ I(0) $\to$ Cointegration exists.

   if test statistic $<$ critical value, reject H_0.

Heteroscedasticity in time series

The **assumption** affected is **t4**, which causes **OLS to no longer be efficient**.
Use tests like Breusch-Pagan or White's, where H_0: No heteroscedasticity. For these tests to work, there should be **no autocorrelation**.

ARCH

An autoregressive conditional heteroscedasticity (ARCH) model is used to analyse dynamic conditional volatility, where the squared errors follow an AR(p) process.
Given the model: $y_t = \beta_0 + \beta_1 z_t + u_t$, with serially uncorrelated errors satisfying $\mathrm{E}(u_t \mid \mathcal{F}_{t-1}) = 0$, an ARCH(1) error variance is specified as:

$$\sigma_t^2 = \mathrm{Var}(u_t \mid \mathcal{F}_{t-1}) = \mathrm{E}(u_t^2 \mid \mathcal{F}_{t-1}) = \alpha_0 + \alpha_1 u_{t-1}^2$$

where $\alpha_0 > 0$ and $0 \leq \alpha_1 < 1$ ensure positivity and covariance stationarity.